

Why Scientist And Partner Co-Development Is Necessary Now

TRANSITION AFTER DECK 1

- The current platform already includes a public route, administration observability, an explainable baseline, and governance surfaces.
- The V2 infrastructure is real: Swiss RAvaFcast research, Himalayan v3 evidence templates, and one-command package triage.
- The science gap is also real: weak layers, regional transfer, quality-controlled labels, station coverage, and promotion evidence are unfinished.
- That gap is exactly where SASE/DGRE partners and the scientist team become central rather than decorative.

READY NOW

Decision-support platform

Public-facing product shell, operator evidence surfaces, and partner-intake templates already exist.

NOT READY NOW

Scientific closure

The next phase is about SASE/DGRE evidence handoff, benchmark ownership, and validation ownership, not declaring victory.

Why This Is Worth The Scientist Team’s Time

OWNERSHIP

Benchmark design authority

The scientist team can define slices, acceptance thresholds, and the cases that matter operationally.

CONTROL

Release-gate veto power

Promotion claims stay blocked until scientist-reviewed evidence says otherwise.

RELEVANCE

Field and weak-layer focus

The hardest real avalanche questions become the center of the roadmap, not an appendix afterthought.

OUTPUT

Publication and pilot value

A co-owned validation pack can become the basis for publishable pilot evidence, not just interface polish.

What makes this scientifically interesting: the stack is inspectable, the claims are gated, and the next step is benchmark ownership rather than UI applause.

Ground Rules: Claim Discipline, Evidence Discipline, Validation Authority

Three rules for the program

Rule 1

Every major statement should map to one of three lanes: live platform, technical evidence, or research agenda.

Rule 2

Validation authority stays with scientists and operators, not with promotional language.

Rule 3

Data-governance and publication expectations should be explicit before deeper pilot work begins.

Consequence

If a gate fails, the claim downgrades immediately instead of being explained away.

WHY SCIENTISTS SHOULD LIKE THIS

- The collaboration asks for challenge and review, not passive endorsement.
- The product team is committing to explicit downgrade rules when evidence is weak.
- The language discipline is part of the operating model, not a one-off deck exercise.

Scientist Role In The Next Phase

BENCHMARK OWNERSHIP

Decide what counts

Choose regions, slices, weak-layer cases, and acceptable benchmark deltas.

EVIDENCE ARBITRATION

Review difficult labels

Resolve ambiguous event records and define what counts as meaningful corroboration.

GOVERNANCE AUTHORITY

Shape promotion criteria

Keep candidate-model promotion tied to scientist-reviewed gates rather than engineering enthusiasm.

PUBLICATION ROLE

Co-author methods if warranted

Use the pilot to generate defensible evidence, not just slideware.

Himalayan v3 Partner Handoff Packet

What partners receive (20+ artifacts, summarized here; full list in the handout)

Start here

README, field dictionary, checksum guide, package index, and source-manifest template.

Evidence to fill

Ten blank v3 CSVs for stations, weather, snowpack, labels, polygons, events, sensing, terrain, reviews, and holdout.

Validation and triage

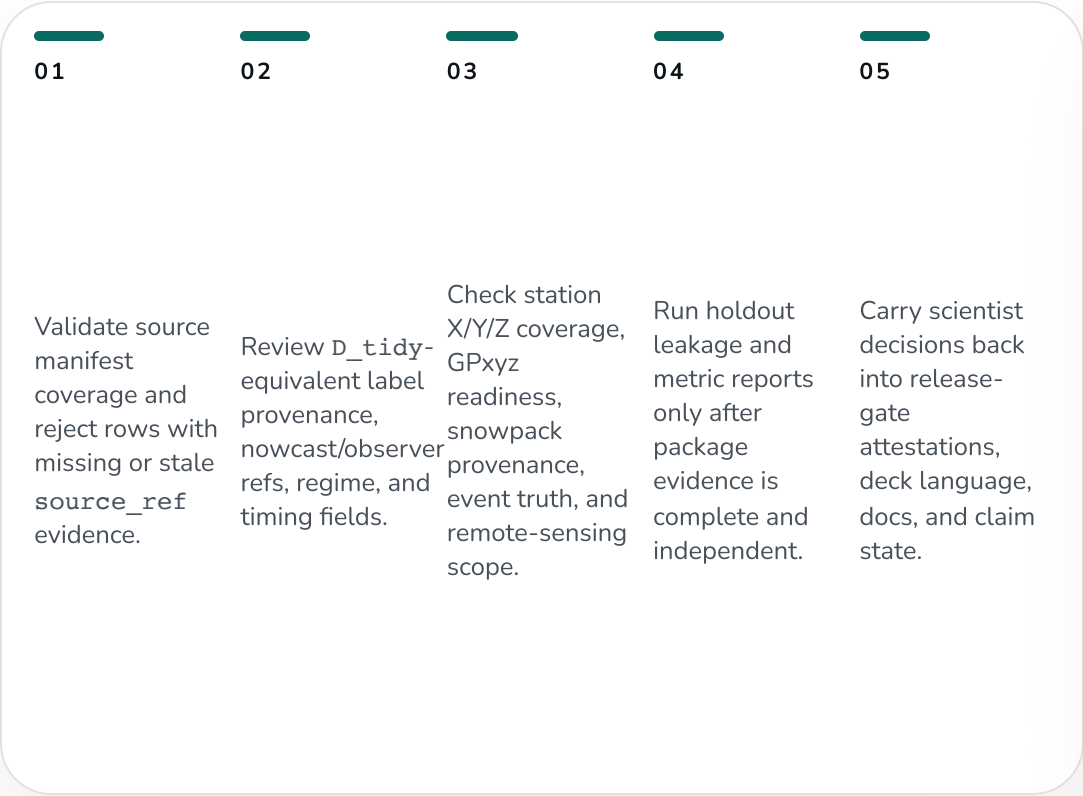
Preflight, evidence validation, source checks, quality score, acceptance checklist, manifest diff, summary, and ledger.

Holdout and claim locks

Pre-registered holdout protocol, leakage audit, metric report, release attestation, and explicit claim gates.

Boundary: blank templates and instructions only. Synthetic fixtures are validator smoke-tests, never partner evidence. See `Scientist_Handout_OnePager.md` for the complete inventory.

Validation Protocol v0 And V3 Triage Loop



WHY THIS MATTERS NOW

- The protocol gives the scientist team and partner reviewers a real review surface from day one.
- It turns “validation” into explicit package, source, label, station, holdout, and release-gate checkpoints.
- It keeps blocked claims blocked until someone can point to the proving artifact and reviewed local evidence.

Critical-Layer And Weak- Layer Program

KNOWN LIMITATION

Proxy wording is not closure

Snowpack proxies and problem framing help, but they do not prove critical-layer fidelity.

SCIENTIST NEED

Weak-layer benchmark slices

The scientist team needs to define the cases that actually matter before promotion is even discussed.

WHY NOW

This is where trust is earned

Senior avalanche scientists will ask this early; answering it openly improves credibility.

Governed Autonomy Roadmap And Promotion Logic

PROMOTION LOGIC

GOVERNED
INGEST

Evidence enters with confidence, source, corroboration, and decay metadata.

BENCHMARK
REVIEW

Scientists inspect narrow evidence mixes, failure slices, and drift signals.

GATE REVIEW

pss_gate_passed, shadow quality, SAR gates, and readiness remain explicit.

PROMOTION
ONLY IF
EARNED

Activation follows artifacts and review, not model novelty alone.

Recommended framing: governed candidate evidence fusion. **Avoid:** finished autonomous avalanche intelligence.

SAR Qualification Path

DATA ACCESS

Reliable scene access

Qualification needs predictable multi-region scene availability and artifact bookkeeping.

LABELS

Reference scarcity

Held-out avalanche labels remain too thin for strong regional promotion claims.

TERRAIN PHYSICS

Shadow and layover

Radar geometry creates real blind spots that need explicit treatment, not optimistic assumptions.

WEATHER REGIME

Dry-snow detectability limits

Some relevant avalanche signatures remain difficult or ambiguous in the current SAR path.

OPERATIONS

Revisit timing

The workflow must respect timing gaps before stronger operational claims are earned.

QUALIFICATION SCOPE

Region by region

SAR should qualify per use case and region rather than via one global promotional statement.

Scientist-In-The-Loop Pilot Design

WEEKS 1-3 · AUDIT + PROTOCOL LOCK	WEEKS 4-6 · EVIDENCE	WEEKS 7-9 · AUDIT + PROTOCOL LOCK	WEEKS 10-12 · WEEK 13+ DECISION ATTESTATION	PER-WEEK HANDOUT
Partner handoff session (W1), discipline + SHA-256 metadata + GPxyz blocked_station	D_tidy label provenance + weak-layer slice definitions (W4), full validator/leakage triage + 100-point score (W5), Stage-1 RF4 feasibility spike + SAR read-only review (W6). Acceptance: ≥1 region scores ≥50/100; SAR not promoted.	Refined-discretization audit (W7), weak-layer evidence pack round 1 (W8), station PRE-REGISTERED local holdout protocol with scientist co-signature (W9). Acceptance: protocol SHA-256 captured, immutable thereafter.	Weak-layer round 2 spike session producing one of: Phase 2 Scoping.md, Phase 2 Pilot Charter.md, or Phase 2 Termination Memo.md. RELEASE-GATE: Scientist Lead can extend, narrow, block, or terminate at any point weekly gate. ATTESTATION OR CLAIM_REVIEW, BLOCKED LEDGER ENTRY (W12) No silent failures.	The week-by-week plan with deliverables, owners, acceptance gates, risk triggers, and contingencies (Swiss-only sandbox, synthetic-pseudo-stations, re-submission cycle, scientist-led narrow pilot) lives in MVP_V2_13_Week_Pilot_Plan.md. Companion artifacts: MVP_V2_Action_List.md and MVP_V2_Weekly_Progress_Template.md.

Success is not “the model looked good.” Success is a benchmark result the scientist team is willing to stand behind narrowly and carefully.

D3 10

Data And Field Requirements

SCIENTIST INPUT

D_tidy-quality danger labels

Separate raw bulletins from reviewed nowcast, observer, event, or reanalysis-backed truth.

FIELD CONTEXT

Station and terrain metadata

GPxyz needs latitude, longitude, elevation, density diagnostics, and region coverage before spatial claims.

EVENT CURATION

Label arbitration and event truth

Ambiguous, duplicated, or thinly sourced avalanche events need shared review and source-manifest backing.

HONEST CAVEAT

Offline report replay is not the same as field reliability

The queue-and-replay mechanism and v3 intake validators exist, but neither proves field-device robustness or Himalayan accuracy.

Engineering And Partner Workstreams Are Already Structured

WORKSTREAM

Claim-state hardening

Every major statement is now mapped to a clear evidence lane and blocked-claim rule.

WORKSTREAM

Partner-package triage

`run_himalayan_partner_package_triage` produces preflight, source, evidence, holdout, ledger, dashboard, manifest, and traceability outputs.

WORKSTREAM

Governed autonomy reframing

The story has shifted from autonomy rhetoric to governed candidate evidence fusion.

WORKSTREAM

Benchmark and protocol packaging

Benchmark pack v0, validation protocol v0, and the v3 handoff packet are ready as discussion starters.

These workstreams improve the next conversation and pilot design. They do not by themselves convert future research into present-day operating readiness.

Team Shape And Collaboration Model

CORE

Full-stack and mapping engineers

Own the public product shell, admin surfaces, map interactions, and evidence display quality.

SCIENCE AND DATA

ML or data engineer plus scientist review

Own benchmark assembly, evaluation slices, governance logic, and failure analysis packaging.

REMOTE SENSING

Geospatial or SAR specialist as needed

Engage selectively when the qualification path moves from schema to held-out regional evidence.

AUTHORITY CENTER

Scientist team at the center

Owns benchmark authority, review cadence, and the final interpretation of what becomes credible enough to promote.

Three-Phase Timeline, Budget, And Infrastructure

PHASE 1 • 0-3 MONTHS Platform hardening Lock claim states, verify core evidence surfaces, and package benchmark intake. Budget: INR 25-40 lakh	PHASE 2 • 3-9 MONTHS Scientist-in-the-loop pilot Scientists co-own benchmark design, acceptance thresholds, and review loops. Budget: INR 40-90 lakh	PHASE 3 • 9-18 MONTHS Validation expansion Qualify SAR candidates, expand multi-region benchmarking, and publish pilot evidence if earned. Budget: INR 1.2-3.0 crore	INFRASTRUCTURE ASSUMPTIONS • Hosted web shell continues on the current React, Supabase, and Netlify baseline. • Modal.com and GPU spend stay tightly bounded to candidate-model and SAR qualification work. • Scientist time is budgeted as validation authority and benchmark ownership, not just occasional advisory review.
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Concrete Ask, Decision Options, And Next Steps

OPTION 1

SASE/DGRE handoff session

- Walk through README, field dictionary, checksum guide, source manifest, and blank v3 CSVs.
- Low commitment, high clarity before data formatting begins.

OPTION 2

90-day evidence pilot

- Partner package submission, triage, scientist review, and go or no-go gates.
- Turns curiosity into inspectable pilot evidence.

OPTION 3

Deeper co-development track

- Commit to validation authority, publication intent, and a longer program arc.
- Best only if the scientist team wants to shape the roadmap materially.